

Oral manifestations of thyroid disorders



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Objectives



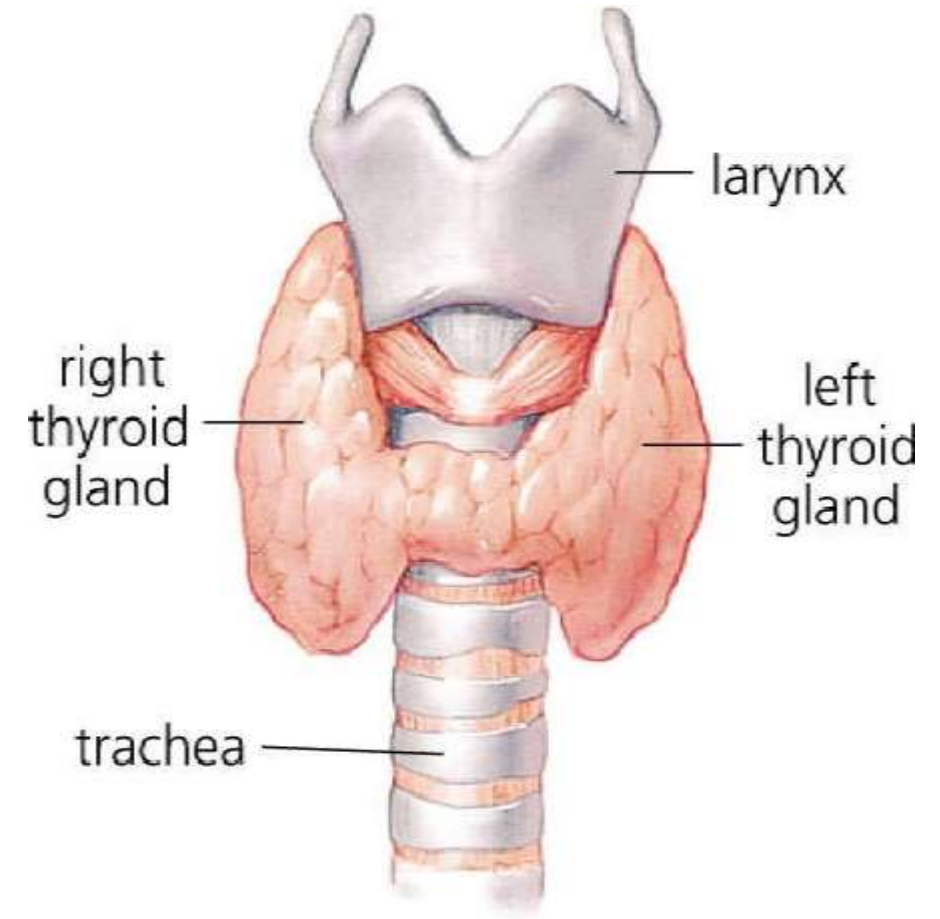
- To understand basic thyroid axis physiology
- To know the common causes of hypo and hyperthyroidism
- To recognise the signs and symptoms associated with hypo and hyperthyroidism
- To understand TFT interpretation
- To know the management for hypo and hyperthyroidism
- Recognize the oral implication of thyroid diseases

INTRODUCTION

Thyroid gland is composed of two elongated lobes on either side of the trachea that are joined by a thin isthmus of thyroid tissue located at or below the level of the thyroid cartilage

It Secretes-

- Thyroxin (T_4)
- Tri-iodothyronine (T_3)
- Calcitonin





Function of Thyroid gland

secretes 3 main hormones

- Thyroxine (T4)
 - Triiodothyronine (T3)
- Energy & Growth
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- Calcitonin → Control of calcium

Hypothalamus-Pituitary-Thyroid Axis

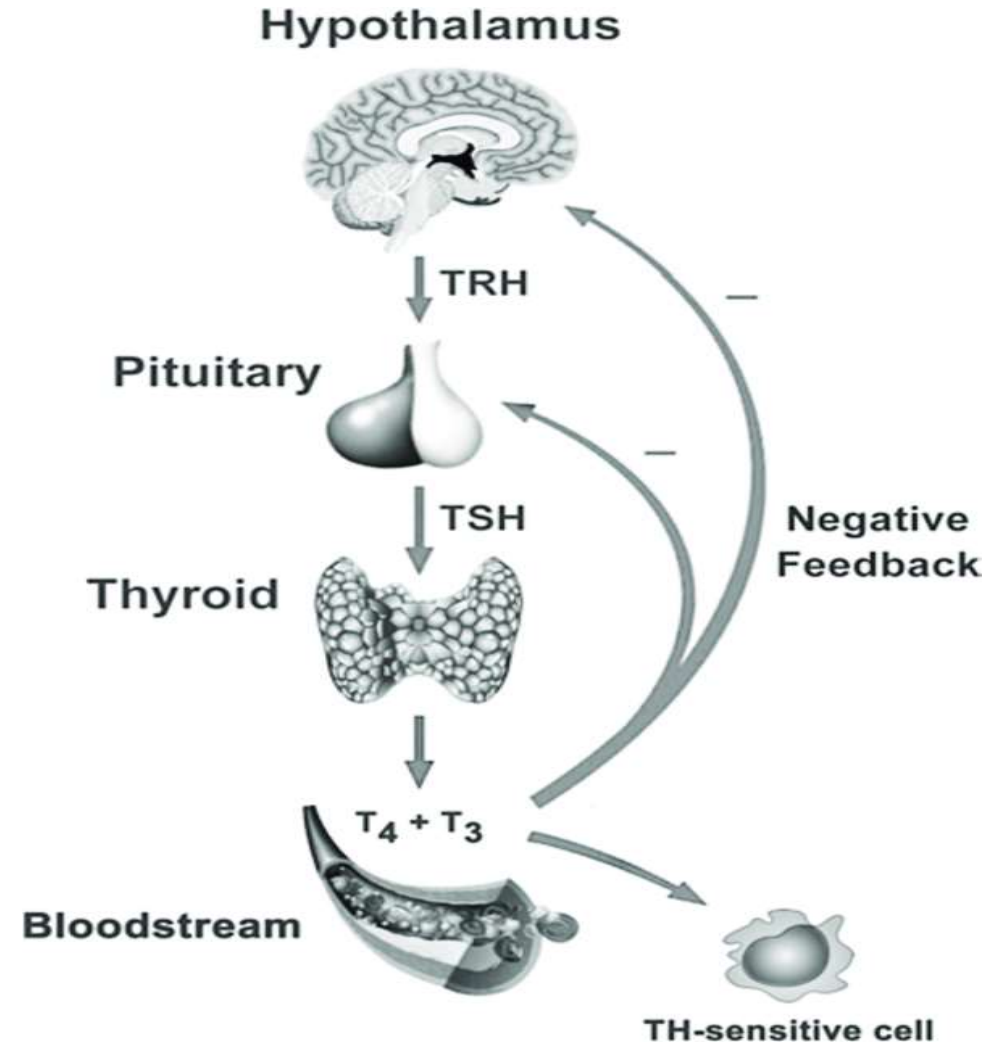


T3 & T4

Facilitate normal growth and development
Increase metabolism
Increase catecholamine effects

TSH

Most useful marker of thyroid hormone function
Released in a pulsatile diurnal



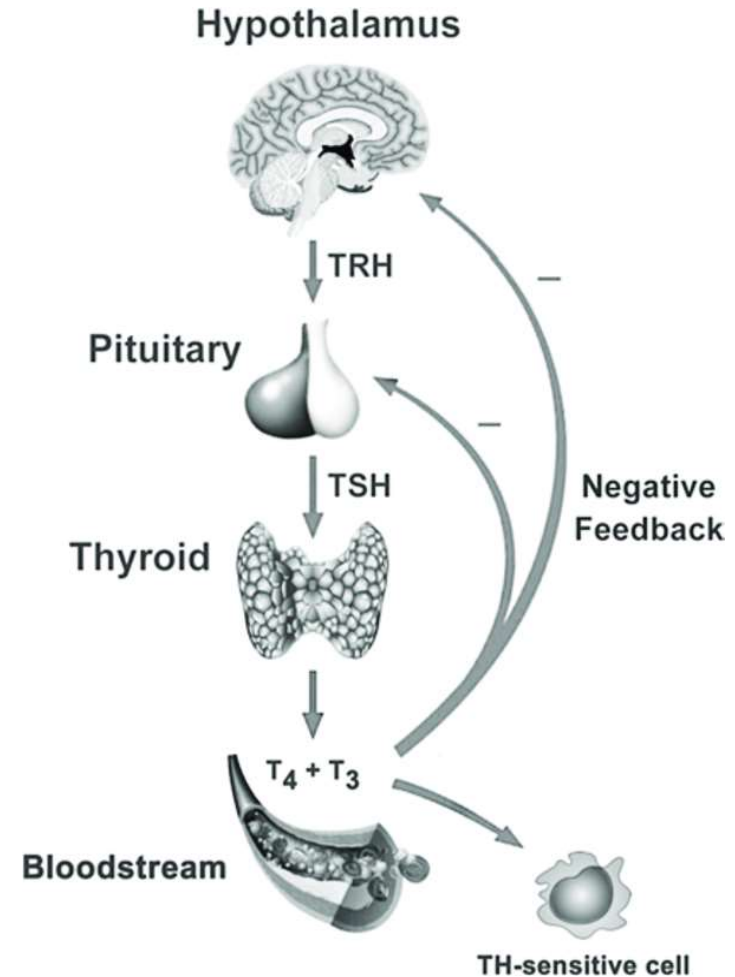
Hypothyroidism



Causes

- **Insufficient thyroid hormone**

1. Primary: thyroid gland failure
2. Secondary: pituitary gland failure
3. Tertiary: hypothalamus failure



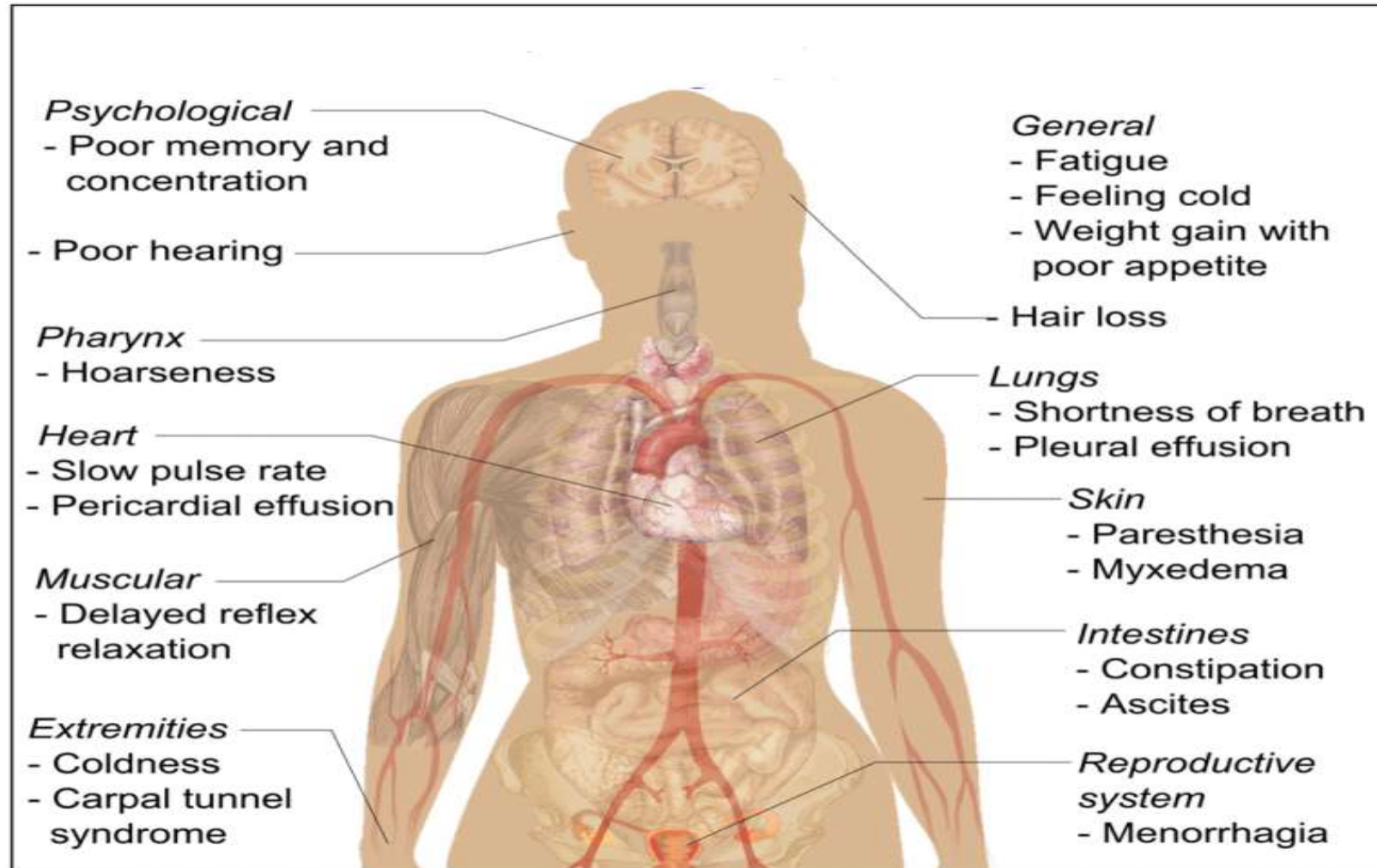
Primary hypothyroidism



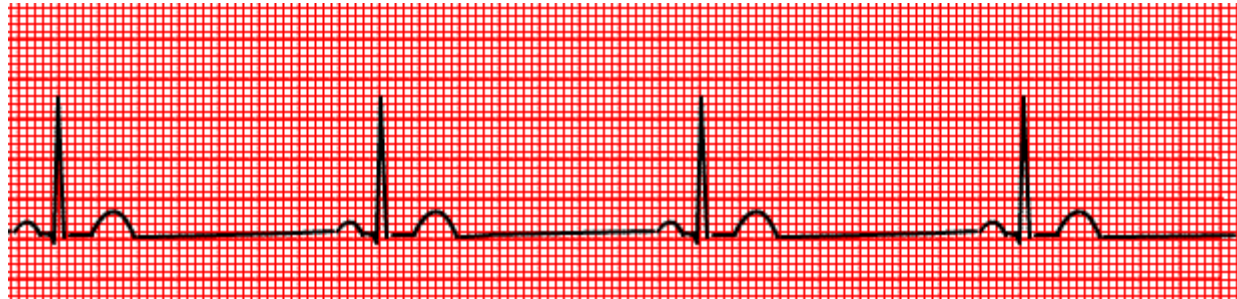
Causes

- Iodine deficiency- most common cause worldwide
- Congenital
- Autoimmune mediated
 - Hashimoto's thyroiditis- B lymphocytes invade thyroid
- Iatrogenic- post-thyroidectomy or radio-iodine treatment
- Drug-induced - Anti-thyroid, lithium, amiodarone
- Severe infection
- Trauma to thyroid/pituitary/hypothalamus
- Pituitary tumour

Signs and symptoms of hypothyroidism



Signs and symptoms of hypothyroidism



Signs and symptoms of hypothyroidism



1. Puffiness of the eye
2. Dry skin
3. Yellow skin
4. Coarse features



Signs and symptoms of hypothyroidism



Signs and symptoms of hypothyroidism



Congenital Hypothyroidism



Hypothyroidism - Management

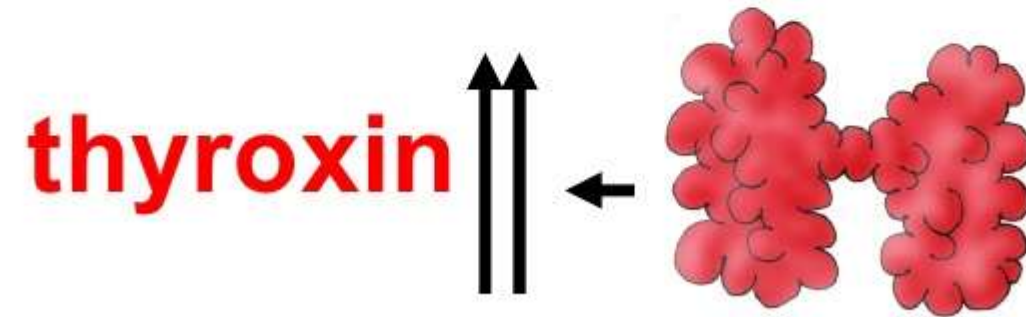


- Conservative
 - Lifestyle - smoking cessation, weight loss
- Medical
 - **Levothyroxine (T4)**
 - Caution in patients with IHD- risk of exacerbation of MI
- Surgical
 - Symptomatic - carpal tunnel decompression, thyroidectomy if compression of local structures

Thyrotoxicosis



Def.: It is an increase in the thyroxine production by the thyroid gland & this is either due to the whole gland enlargement or a thyroid nodule



Hyperthyroidism Causes

Hyperthyroidism (thyrotoxicosis) is excess thyroid hormone

- **Autoimmune**

- Graves Disease (76%)
 - F>M, age 20-40
 - IgG auto antibodies bind TSH receptors T3 & T4
 - Leads to gland hyper function

- **Toxic adenoma and toxic multinodular goitre**

- **Viral Thyroiditis** (de Quervain's)

- Fever and ESR- self limiting

- **Exogenous Iodine**

- Neonatal thyrotoxicosis

- **Drugs- Amiodarone**

- TSH secreting pituitary adenoma (rare)

- HCG producing tumour

What happens?

An increase in thyroxin secretion.

Increase of response of the body cells to adrenaline.

So signs & symptoms of the disease occur.



7-osteoporosis.



5-Polyurea & diahrea



3- thyroid paradox:
Decrease in weight
although increase
in appetite.



8- Thyrotoxic goiter



6-Fatigability



4- increase sweating & sense
of hotness in weather





- **Exophthalmos and eye congestion**
- **Lid retraction**



Figure 5

– **Exophthalmos and Staring look**

Clinical features of hyperthyroidism and hypothyroidism

Hyperthyroidism

Anxiety

Restlessness

Tachycardia

Weight loss

Thinning skin

Heat intolerance

Frequent bowel movements

Goitre possible

Low TSH

High T3/T4

Hypothyroidism

Depression

Fatigue

Bradycardia

Weight gain

Dry, itchy skin

Cold intolerance

Constipation

Goitre possible

High TSH

Low T3/T4

Hyperthyroidism - Management



- Conservative

- Smoking cessation – especially with Graves's ophthalmology, associated with worse prognosis

- Medical

- Symptomatic – β -blockers
- Carbimazole, propylthiouracil (50% relapse)
 - Risk of agranulocytosis
- Radio-iodine treatment – avoid contact with pregnant women and small children

Investigating Thyroid Disease



- TSH- first thing should be assessed
 - Normal range 0.5-5 $\mu\text{U/ml}$
 - Suppressed= Hyperthyroid
 - Elevated= Hypothyroid

If TSH abnormal request Free T4

- Elevated= Hyperthyroid
- Suppressed= Hypothyroid

Enlarged Thyroid gland (Goiter)



- Iodine deficiency , Excess iodine
- Autoimmune thyroiditis
lithium ingestion, which decrease release of thyroid hormone
- Exposure to radiation
- Infectious agents
- Thyroid malignancy



Oral manifestations of thyroid disorders



Hypothyroidism	Hyperthyroidism
• Salivary gland enlargement • Compromised periodontal health – delayed bone resorption • Macroglossia • Glossitis • Dysgeusia • Delayed dental eruption • Enamel hypoplasia in both dentitions, (being less intense in the permanent dentition) • Anterior open bite • Micrognathia • Thick lips • Mouth breathing	• Increased susceptibility to caries • Increased susceptibility to periodontal disease • Enlargement of extraglandular thyroid tissue • Burning mouth syndrome • Accelerated dental eruption • Maxillary and mandibular osteoporosis • Development of connective tissue diseases like Sjogren's syndrome or Systemic lupus erythematosus

Oral manifestations of thyroid disorders

Recognition of the signs and symptoms of hyperthyroidism is very important in dentistry.

The elevated levels of thyroid hormones in these patients make them extremely sensitive to sympathomimetic drugs such as epinephrine, which is used in dentistry as a vasoconstrictor in local anesthesia

Administration of sympathomimetics can cause **hypertensive crisis, tachycardia, and/or dysrhythmia.**

Dental treatment consideration in thyroid disease



Hemostasis

Patients with high blood pressures may require increased attention and a longer duration of local pressure to stop bleeding.

Anti-thyroid drugs namely propylthiouracil (PTU) has anti-vitamin K activity and can cause hypoprothrombinemia and bleeding that poses a risk for hemorrhage. Thus, patients taking PTU must be carefully evaluated before surgery or invasive dental treatment

Dental treatment consideration in thyroid disease



Susceptibility to infection

Thionamides may cause a very rare reaction of agranulocytosis (0.5% of patients) that can result in oral infections and inadequate wound healing. These post-operative complications could be prevented if clinicians carefully follow precautions stated with thionamide

Drug actions and interactions

Combination analgesics containing acetylsalicylic acid (ASA) are contraindicated in patient with hyperthyroidism because ASA interferes with the protein binding of T4 and T3, thereby increasing their free form. This may worsen the symptoms of thyrotoxicosis



Oral manifestations of thyroid disorders

The hyperthyroid patients` increased sensitivity to circulating catecholamines, They may be resistant to the effects of CNS depressants such as diazepam and other anti-anxiety drugs.

Benzodiazepines, barbiturates, and narcotic analgesics may cause respiratory and cardiac depression in patients with hypothyroidism.

The L-thyroxine (T4) hormone replacement drug can be synergistic with sympathomimetic drugs used in dentistry.

DENTAL treatment consideration in thyroid diseases



Hyperthyroidism	Hypothyroidism
• Avoid the use of adrenaline and control the spread of Infection • Patients treated with propylthiouracil, a complete blood count should be done • NSAIDs and aspirin should be used with caution • Treatment should be discontinued if signs or symptoms of a thyrotoxic crisis develop. • These patients are susceptible to central nervous system depressant drugs such as barbiturates • These patients are vulnerable to cardiovascular disease; a proper blood investigations and consultation with the physician is needed. • Management of stress is important in these patients.	• A complete blood count before performing dental surgery is recommended. • Avoid the use of adrenaline, control oral infection and do not use central nervous depressant drugs. • Dentist should have knowledge of drug interactions of thyroxine. • If myxedematous coma develops, dentist should avoid any planned dental surgery. • Patients are vulnerable to cardiovascular disease; proper consultation with the cardiologist is required before any surgical intervention. • Good glycemic control is required in patients with diabetes mellitus who will be undergoing a dental surgical treatment.

Case Discussion

History: A 40-year-female complains of progressive weight gain of 10 kg in 1 year, fatigue, postural dizziness, loss of memory, slow speech, deepening of her voice, dry skin, constipation, and cold intolerance.

Physical examination: Vital signs include a temperature 36.5, pulse 58/minute and regular, BP 110/60. She is moderately obese and speaks slowly and has a puffy face, with pale, cool, dry, and thick skin. The thyroid gland is not palpable. The deep tendon reflex time is delayed.

- **Laboratory studies:** CBC and differential WBC are normal. The serum T4 concentration is 3.8 ug/dl (N=4.5-12.5), the serum TSH is 1 uU/ml (N=0.2-3.5), and the serum cholesterol is 255 mg/dl (N<200).

1. What is the likely diagnosis?
2. What symptoms suggest this diagnosis?
3. What physical signs support it?
4. Which lab results support it?
5. What are the possible causes?
6. What tests can confirm the diagnosis?

Case Discussion

Patient: 48-year-old female

History: Chronic fatigue, weight gain, constipation, dry skin

Oral Findings: Macroglossia, delayed wound healing, and thickened lips

Diagnosis: Primary hypothyroidism (confirmed by elevated TSH, low T3/T4)

□ Dental Relevance:

- **Macroglossia** can cause speech issues and may affect denture fitting.
- **Delayed wound healing** could complicate extractions or periodontal therapy.
- **Increased bleeding tendency** due to altered metabolism.
- **Avoid CNS depressants** (like sedatives) as they may

Case Discussion

Patient: 26-year-old female

History: Palpitations, anxiety, heat intolerance, weight loss

Oral Findings: Accelerated dental eruption, tremors during procedures, possible osteoporosis on radiographs

Diagnosis: Graves' disease (low TSH, high T3/T4, positive anti-TSH receptor antibodies)

☐ **Dental Relevance:**

- **Increased sensitivity to epinephrine** — avoid vasoconstrictors in local anesthesia.
- **Higher risk of thyroid storm** during stress (e.g., dental surgery) .

Case Discussion

Patient: 7-year-old child

History: Developmental delay, short stature, slow speech

Oral Findings: Delayed tooth eruption, malocclusion, macroglossia

Diagnosis: Congenital hypothyroidism (confirmed at birth, on levothyroxine)

☐ **Dental Relevance:**

- **Delayed eruption and dental development** require early orthodontic referral.
- **Macroglossia and malocclusion** impact function and aesthetics.